



A Two-stage AI/ML Framework Integrating Ayurvedic Phenotyping and Clinical Data for Early Alzheimer's Risk Detection

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KEY TAKE AWAY

A culturally contextualized neuro-integrative AI/ML framework combining Ayurvedic phenotyping, clinical screening and digital biomarkers shows strong feasibility for Stage-0 dementia risk stratification in low-resource settings.



Integrative Design:
Combines Ayurvedic phenotyping, clinical biomarkers, and digital health signals.



Scalable Screening:
Demonstrates strong technical feasibility for Stage-0 dementia risk stratification.



LC-Centric Perspective:
Focus on early vulnerability of the Locus Coeruleus (LC)—one of the first regions affected in AD.



Target Audience:
Optimized as a low-cost screening pipeline for resource-limited settings.

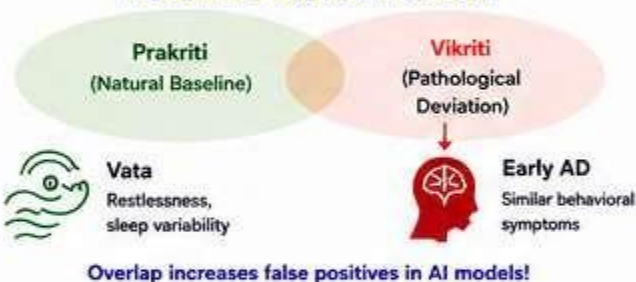


System Scope:
Classifies early risk profiles; not intended as a final clinical diagnostic tool.

1. INTRODUCTION & RESEARCH CONTEXT

- Early detection of Alzheimer's disease remains difficult in South Asian populations due to limited access to advanced diagnostics and late presentation of cognitive symptoms.
- Ayurveda offers long-standing diagnostic frameworks such as Prakriti-based constitutional profiling and Nadi (pulse) assessment, which capture subtle physiological and behavioral variations that may precede clinical impairment.
- However, these traditional insights have rarely been integrated with modern clinical assessment using AI/ML approaches.
- This work presents a critical gap in developing accessible, culturally grounded tools for early risk stratification of Alzheimer's and related neurodegenerative conditions.

PHENOTYPIC OVERLAP PROBLEM



2. IDENTIFIED RESEARCH GAP

Current systems cannot clearly distinguish between:



→ Leads to false positives and inaccurate early risk prediction.

Core Challenge:

Lack of baseline personalization, multimodal fusion, and privacy-preserving learning in early screening tools.

Research Gap:

Need for an integrative, scalable, culturally contextualized framework that fuses Ayurvedic phenotyping with clinical and digital biomarkers for Stage-0 dementia risk detection.

3. TRI-LAYER VALIDATION FRAMEWORK



Layer 1: Ayurvedic Baseline Assessment (Implemented)

- Questionnaire-based phenotyping
- Prakriti classification
- Baseline physiological profiling



Layer 2: Clinical Risk Fusion (Implemented)

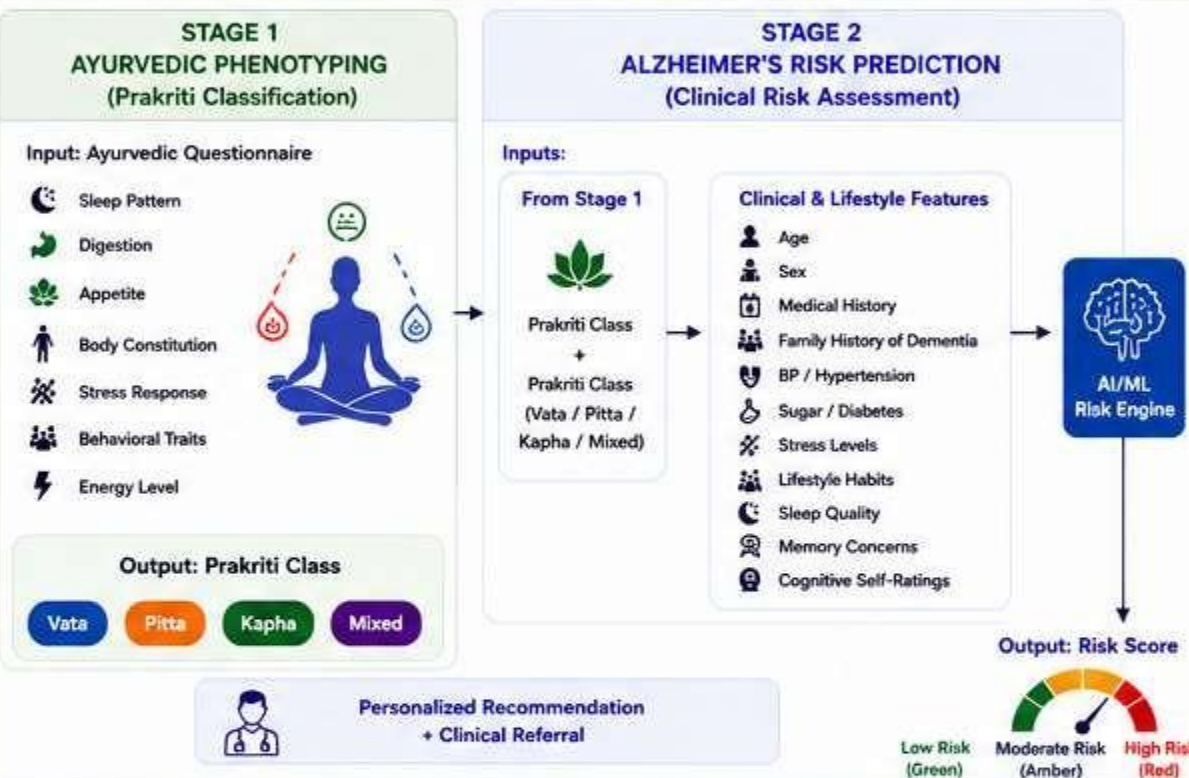
- Fuse medical history, lifestyle and metabolic indicators with Prakriti features
- Improve personalized risk stratification



Layer 3: Temporal + Digital Biomarker Validation (Future)

- 16-30 day longitudinal monitoring
- HRV, sleep tracking, wearable biomarkers
- Federated learning for privacy preservation

4. TWO-STAGE AI/ML PIPELINE (CURRENT PROTOTYPE)



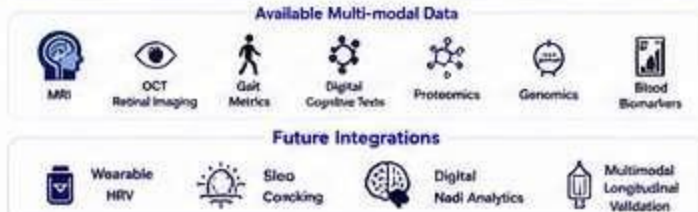
5. MOLECULAR MECHANISM OF FEMALE-BIASED LC VULNERABILITY



Clinical Relevance: Sex should be treated as a highly weighted Stage-2 predictive feature to avoid underestimating early female brainstem-driven tauopathy risk.

8. FUTURE WORK & EXTERNAL VALIDATION

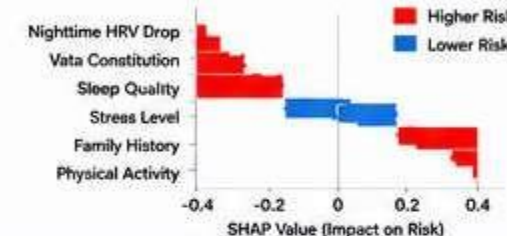
External validation using IISc Centre for Brain Research cohorts, in partnership with AD Data Initiative & Microsoft Research India.



6. EXPLAINABLE AI (XAI) LAYER

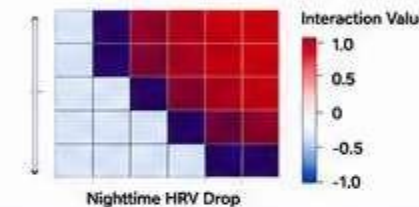
Unified Local SHAP Attribution

Converts Prakriti baseline, digital biomarkers and lifestyle metrics into standardized Shapley values quantifying how each feature pushes risk higher (↑) or lower (↓).



SHAP Interaction Analysis

Reveals non-linear interactions between features.
Vata Baseline × Nighttime HRV Drop → Risk Spike



Risk Waterfall Visualization

Transparent view of baseline risk, push factors, protective factors and final risk.



7. RESULTS – PROTOTYPE VALIDATION (SYNTHETIC DATA)

- ✓ Successful end-to-end execution of two-stage pipeline
- ✓ Stable risk stratification across simulated cohorts
- ✓ Prakriti features improved segmentation vs clinical-only baseline
- ✓ Tri-layer validation reduced false positives from overlap
- ✓ XAI layer identified dominant risk contributors
- ✓ Strong architectural feasibility and logical consistency

NOTE: Synthetic validation confirms feasibility; real-world clinical validation remains future work.

9. CONCLUSION

This work presents a novel neuro-integrative AI/ML framework fusing Ayurveda (IKS) and modern clinical assessment for early dementia risk screening. By combining constitutional phenotyping, clinical biomarkers, digital health signals and explainable AI, the framework offers a scalable, culturally contextualized and low-cost solution for Stage-0 dementia screening in resource-limited settings.



10. REFERENCES

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- Dagum P. (2018) Digital Biomarkers of Cognitive Function
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11. CONTACT

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